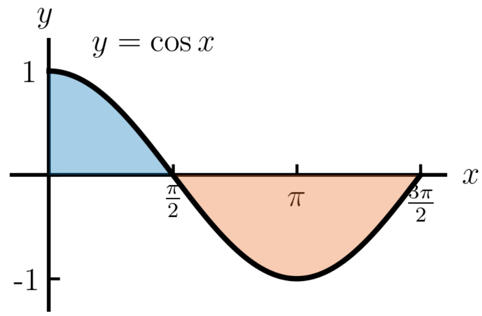
The Definite Integral

In this lesson, we will explore net area bounded by a curve, define the definite integral, and practice matching definite integrals to limits of Riemann sums.

So far, we have approximated the area under a non-negative function f on $[a, b]$. If f takes positive and negative values, then Riemann sums approximate the **net area** bounded by the graph of f and the x-axis.

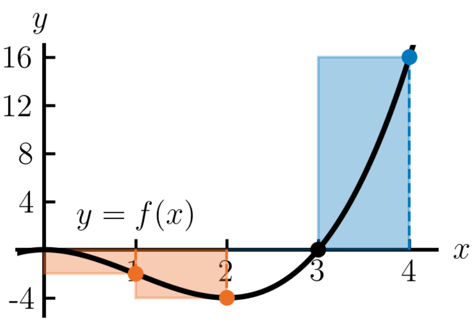


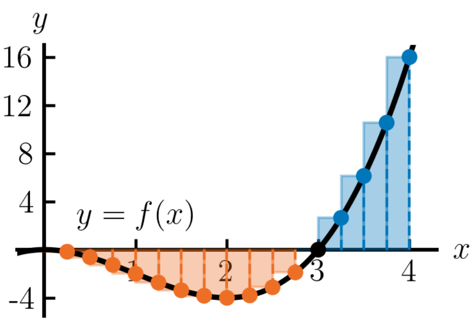
Definition Net Area: The net area bounded by the graph of f and the x-axis on $[a, b]$ is the area above the x-axis minus the area below the x-axis.

Example 1: Let $f(x) = x^3 - 3x^2$. Approximate the net area bounded by the graph of f and the x-axis on the interval $[0, 4]$ using 4 right-endpoint rectangles.

Solution: On $[0, 4]$ with $n = 4$ rectangles, the width is

The right endpoints are





The Riemann sum $\sum_{k=1}^{n} f(x_k^*)\Delta x$ approximates the net area bounded by the graph of f and the x-axis on $[a, b]$. Taking the limit as $n \to \infty$, the Riemann sum approaches the exact net area.

Definition **Definite Integral:** The definite integral of f on $[a, b]$ is defined by

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x,$$

if the limit exists and is independent of the choice of x_k^*. If the definite integral exists, we say that f is **integrable** on $[a, b]$.

- a and b are the **limits of integration**
- $f(x)$ is the **integrand**
- x is the **variable of integration**

Theorem: If f is continuous on $[a, b]$, or is bounded with finitely many discontinuities on $[a, b]$, then f is integrable on $[a, b]$.

Example 2: The expression $\lim_{n\to\infty} \sum_{k=1}^{n} \left((x_k^*)^2 + 5x_k^* - 1\right) \cdot \frac{3}{n}$ is a limit of a Riemann sum for a function f on $[0, 3]$. Find the function f and write the expression as a definite integral.

In the last example, we wrote a limit of a Riemann sum as a definite integral. Next, try this interactive problem on the website to go in the other direction: writing a definite integral as a limit of a Riemann sum.

Interactive Problem 1: Write the definite integral $\int_1^4 \sqrt{x}\, dx$ as a limit of a right-endpoint Riemann sum.

(a) The definition of the definite integral is $\int_a^b f(x)\, dx =$

(b) The width of each subinterval is $\Delta x =$

(c) The formula for right-endpoints is $x_k^* =$

(d) The right-endpoints evaluate to $x_k^* =$

(e) Putting this all together,

$$\int_1^4 \sqrt{x}\, dx =$$